

Feibelman d-parameters

This set of notes follows the following papers:

Quantum Corrections in Plasmonics and Plasmon–Emitter Interactions

(https://link.springer.com/chapter/10.1007/978-3-030-38291-9_8) (**Reference 1**)

A general theoretical and experimental framework for nanoscale electromagnetism

(<https://www.nature.com/articles/s41586-019-1803-1#MOESM1>) (**Reference 2**)

and, finally,

Plasmon–emitter interactions at the nanoscale

(<https://www.nature.com/articles/s41467-019-13820-z#MOESM2>) (**Reference 3**)

We have a new set of boundary conditions,

$$\mathbf{E}_{2,\parallel} - \mathbf{E}_{1,\parallel} = -d_{\perp} \nabla_{\parallel} (E_{2,\perp} - E_{1,\perp})$$

$$\mathbf{H}_{2,\parallel} - \mathbf{H}_{1,\parallel} = i\omega d_{\parallel} (\mathbf{D}_{2,\parallel} - \mathbf{D}_{1,\parallel}) \times \mathbf{e}_n$$

Reflection and transmission coefficients with d-parameters

We take dielectric in $z > 0$ and metal in $z < 0$. We send in a wave in the dielectric and measure the reflected wave.

For $z > 0$, we have $\mathbf{H}(\mathbf{r}, \omega) = e^{iqx} (e^{-ik_d z} + r e^{ik_d z}) \mathbf{e}_y$ (which is clearly transverse-magnetic polarized) and

$$\nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega \mathbf{D}(\mathbf{r}, \omega) \rightarrow \mathbf{E}(\mathbf{r}, \omega) = (i/\omega \epsilon_0 \epsilon_d) \nabla \times \mathbf{H}(\mathbf{r}, \omega) = - (1/\omega \epsilon_0 \epsilon_d) e^{iqx} \left(k_d (e^{-ik_d z} - r e^{ik_d z}), 0, q (e^{-ik_d z} + r e^{ik_d z}) \right)$$

For $z < 0$, we have $\mathbf{H}(\mathbf{r}, \omega) = t e^{iqx} e^{-ik_m z} \mathbf{e}_y$, where

$k_m^2 = \epsilon_m (\omega/c)^2 - q^2$, $k_d^2 = \epsilon_d (\omega/c)^2 - q^2$. Therefore, our electric field in the metal is given by:

$$\mathbf{E} = -(1/\omega\epsilon_0\epsilon_m)e^{iqx}\left(tk_me^{-ik_mz}, 0, tqe^{-ik_mz}\right)$$

Therefore, the boundary conditions give us:

$$\left[k_d(1-r)/\epsilon_d - k_mt/\epsilon_m\right] = -i(d_\perp q^2)\left[(1+r)/\epsilon_d - t/\epsilon_m\right]$$

(boundary condition on the tangential component of the electric field)

And

$$(1+r) - t = i\omega d_{||}(k_d(1-r) - k_mt)(\mathbf{e}_x \times \mathbf{e}_z) \cdot \mathbf{e}_y(-1/\omega) =$$

$$id_{||}(k_d(1-r) - k_mt)$$

(boundary condition on tangential component of the magnetic field)

We note that $\mathbf{D}$ has both $\mathbf{e}_x$ and $\mathbf{e}_z$ components, but since $\mathbf{e}_n = \mathbf{e}_z$ and $\mathbf{e}_z \times \mathbf{e}_z = 0$, only the $\mathbf{e}_x$ component of the displacement field contributes in the second boundary condition. We solve this linear system of equations. Solving for t from the second equation gives us:

$$t(1 - id_{||}k_m) = (1 - id_{||}k_d) + r(1 + id_{||}k_d)$$

We rewrite this as:

$$t = \frac{1 - id_{||}k_d}{1 - id_{||}k_m} + r \frac{1 + id_{||}k_d}{1 - id_{||}k_m}$$

At this point, to ensure we have not made any algebraic mistakes, let's first verify that we get the correct relation if $d_{||} = d_\perp = 0$. In this case, we obtain:

$$k_d(1-r)/\epsilon_d - (1+r)k_m/\epsilon_m = 0 \rightarrow \frac{k_d\epsilon_m - k_m\epsilon_d}{\epsilon_d\epsilon_m} = r \left[\frac{k_d\epsilon_m + k_m\epsilon_d}{\epsilon_d\epsilon_m} \right]$$

Which gives us:

$$r = \frac{k_d\epsilon_m - k_m\epsilon_d}{k_d\epsilon_m + k_m\epsilon_d},$$

which is correct (and may be compared against **Equation 1a** of **Reference 3** in the limit $d_{\perp}, d_{\parallel} \rightarrow 0$).

Satisfied that we have not done anything stupid (yet), we derive the Fresnel coefficients with nonlocal corrections included. We write the first boundary condition below:

$$\left[k_d(1-r)\varepsilon_m - k_m t \varepsilon_d \right] = -i(d_{\perp} q^2) \left[(1+r)\varepsilon_m - t \varepsilon_d \right]$$

We multiply both sides of this equation with $1 - i d_{\parallel} k_m$ and we define, for convenience, $t' = (1 - i d_{\parallel} k_m) t$

This then gives us:

$$\left[k_d \varepsilon_m (1-r)(1 - i d_{\parallel} k_m) - k_m \varepsilon_d t' \right] = -i(d_{\perp} q^2) \left[(1+r)\varepsilon_m (1 - i d_{\parallel} k_m) - \varepsilon_d t' \right]$$

We collect all terms proportional to r to the left hand side of this equation:

$$\begin{aligned} -k_d \varepsilon_m + i d_{\parallel} k_d k_m \varepsilon_m - k_m \varepsilon_d - i d_{\parallel} \varepsilon_d k_m k_d + i q^2 d_{\perp} \varepsilon_m - i q^2 d_{\perp} \varepsilon_d = \\ - \left[k_d \varepsilon_m + k_m \varepsilon_d - (\varepsilon_m - \varepsilon_d) (i d_{\parallel} k_d k_m + i q^2 d_{\perp}) \right] \end{aligned}$$

Note that we have ignored terms such as $d_{\perp} d_{\parallel} q^2 k_m \varepsilon_m$, since these are higher order.

We similarly collect all the constant terms on the right hand side of the equation:

$$\begin{aligned} -i d_{\perp} q^2 \varepsilon_m + i d_{\perp} q^2 \varepsilon_d - k_d \varepsilon_m + k_d k_m \varepsilon_m i d_{\parallel} + k_m \varepsilon_d - i d_{\parallel} k_d k_m \varepsilon_d = \\ - \left[k_d \varepsilon_m - k_m \varepsilon_d + (\varepsilon_m - \varepsilon_d) (i q^2 d_{\perp} - i d_{\parallel} k_d k_m) \right] \end{aligned}$$

So we obtain the Fresnel reflection coefficient in the presence of Feibelman parameter corrections:

$$r = \frac{k_d \varepsilon_m - k_m \varepsilon_d + (\varepsilon_m - \varepsilon_d)(iq^2 d_{\perp} - id_{\parallel} k_d k_m)}{k_d \varepsilon_m + k_m \varepsilon_d - (\varepsilon_m - \varepsilon_d)(id_{\parallel} k_d k_m + iq^2 d_{\perp})},$$

which agrees with **Equation 1a** of **Reference 3**.